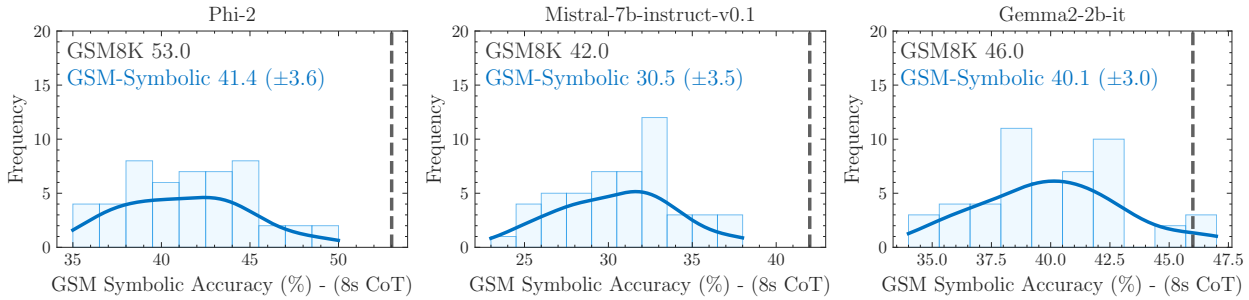
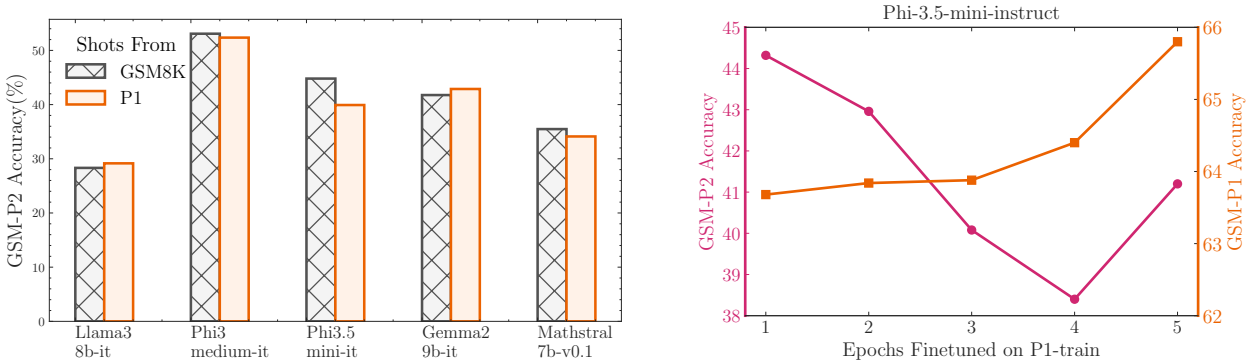


Figure 10: Additional results on performance variation on `GSM-Symbolic`.

Moreover, in Fig. 11b, we demonstrate that fine-tuning Phi-3.5 on `GSM-P1` slightly improves performance on `GSM-P1` while decreasing performance on `GSM-P2`. We have used a set of 50 templates from `GSM-P1`, separate from the test templates, and generated 10000 examples for finetuning training set. Overall, while this direction warrants further research, current results suggest that scaling training data will not be helpful in improving the reasoning capabilities of language models.



(a) (b)

Figure 11: Using in-context shots or finetuning on `GSM-P1` does not improve performance on `GSM-P2`: (a) Compared to the case where 8 shots come from `GSM8K`, when we include shots from `GSM-P1`the performance on `GSM-P2` does not improve. (b) Finetuning on `GSM-P1` can improve performance on `GSM-P1` but not on `GSM-P2`.

A.5 Results on o1-preview and o1-mini

The recently released o1-preview and o1-mini models (OpenAI, 2024) have demonstrated strong performance on various reasoning and knowledge-based benchmarks. As observed in Tab. 1, the mean of their performance distribution is significantly higher than that of other open models.
In Fig. 12 (top), we illustrate that both models exhibit non-negligible performance variation. When the difficulty level is altered, o1-mini follows a similar pattern to other open models: as the difficulty increases, performance decreases and variance increases.
The o1-preview model demonstrates robust performance across all levels of difficulty, as indicated by the closeness of all distributions. However, it is important to note that both o1-preview and o1-mini experience a significant performance drop on `GSM-NoOp` . In Fig. 13, we illustrate that o1-preview struggles with understanding mathematical concepts, naively applying the 10% inflation discussed in

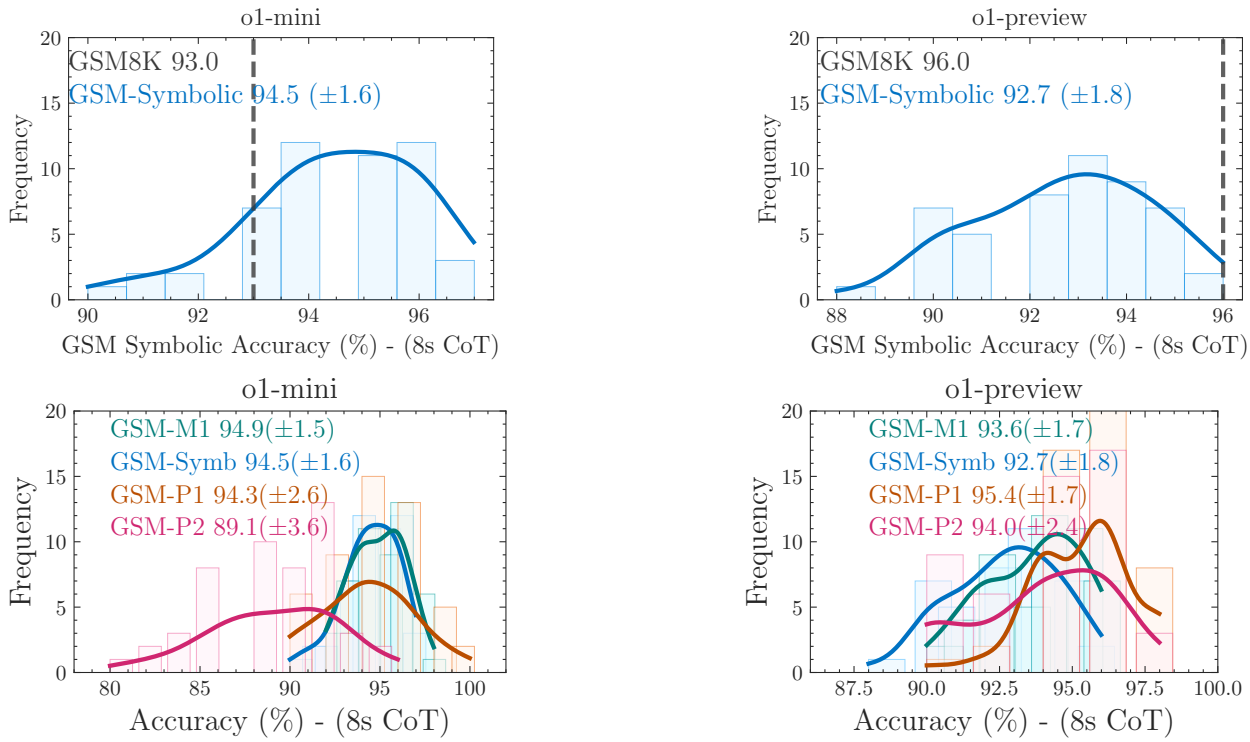


Figure 12: Results on o1-mini and o1-preview: both models mostly follow the same trend we presented in the main text. However, o1-preview shows very strong results on all levels of difficulty as all distributions are close to each other.

the question, despite it being irrelevant since the prices pertain to this year. Additionally, in Fig. 14, we present another example highlighting this issue.

Overall, while o1-preview and o1-mini exhibit significantly stronger results compared to current open models—potentially due to improved training data and post-training procedures—they still share similar limitations with the open models.

GSM-NoOp

Liam wants to buy some school supplies. He buys 24 erasers that now cost $6.75 each, 10 notebooks that now cost $11.0 each, and a ream of bond paper that now costs $19. How much should Liam pay now, assuming that due to inflation, prices were 10% cheaper last year?

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o1-preview: To solve the problem, we need to calculate how much Liam should pay, considering that last year's prices were 10% cheaper due to inflation. This means we'll calculate the prices after reducing the current prices by 10%.

Step 1: Calculate last year's prices by reducing the current prices by 10%.
- Eraser price last year: $\$6.75 \times 0.90 = \$6.075$
- Notebook price last year: $\$11.00 \times 0.90 = \$9.90$
- Bond paper price last year: $\$19.00 \times 0.90 = \$17.10$

Step 2: Calculate the total cost using last year's prices.
- Total cost for erasers: 24 erasers $\times \$6.075 = \$145.80$
- Total cost for notebooks: 10 notebooks $\times \$9.90 = \$99.00$
- Total cost for bond paper: 1 ream $\times \$17.10 = \$17.10$

Step 3: Sum up all the costs to get the total amount Liam should pay.
$\$145.80 + \$99.00 + \$17.10 = \$261.90$

Answer:$261.90

Figure 13: Sample response from o1-preview on an example from GSM-NoOp: the model blindly applies the inflation rate, even though the inflation amount is irrelevant as the question clearly indicates the given prices are for "now" and not last year.